

Theory note S2 — The graph Laplacian, algebraic connectivity λ_2 , and the link range under partial-band jamming

Chang Young Ham

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Purpose. S2 summarises a whole network by one number, the algebraic connectivity λ_2 [1]. This note proves the two facts the simulator relies on ($\lambda_2 > 0$ iff connected; $\lambda_2 \leq \frac{N}{N-1}\delta_{\min}$), states what each verification check tests, shows that in this model a link is up exactly when it is shorter than a *link range*, and uses that to explain the two sweeps. Mathematics used: symmetric matrices, eigenvalues, quadratic forms, the Courant–Fischer characterisation.

1 From links to a matrix

Take N nodes. From S1, each pair (i, j) has a bit error rate $P_b(d_{ij}, J/S, \rho)$; the link is *up* if $P_b \leq \text{BER}_{\text{out}}$. Collect the result in the adjacency matrix

$$A_{ij} = \begin{cases} 1 & \text{link } (i, j) \text{ up, } i \neq j \\ 0 & \text{otherwise,} \end{cases}$$

symmetric with zero diagonal. The degree $d_i = \sum_j A_{ij}$ counts a node’s live links; $D = \text{diag}(d_1, \dots, d_N)$. The *graph Laplacian* is

$$L = D - A.$$

Its rows sum to zero by construction: $L\mathbf{1} = D\mathbf{1} - A\mathbf{1} = (d_i) - (d_i) = 0$.

2 Three facts about L

(a) **L is symmetric, so its eigenvalues are real.** D and A are symmetric. Hence L has N real eigenvalues, which we order $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$, and an orthonormal eigenbasis. `np.linalg.eigvalsh` is the solver for this case (the “h” is for Hermitian).

(b) **The quadratic form is a sum of squared differences over edges.** For any vector $\mathbf{x} \in \mathbb{R}^N$,

$$\mathbf{x}^\top L \mathbf{x} = \sum_i d_i x_i^2 - \sum_{i,j} A_{ij} x_i x_j = \sum_{\{i,j\} \in E} (x_i - x_j)^2 \geq 0. \quad (1)$$

To see the second equality, expand $\sum_{\{i,j\} \in E} (x_i - x_j)^2$: each edge contributes $x_i^2 + x_j^2$ (which sums to $\sum_i d_i x_i^2$ over all edges touching i) and $-2x_i x_j$ (which sums to $-\sum_{i,j} A_{ij} x_i x_j$ since each unordered edge appears twice in the double sum).

(c) **$\lambda_1 = 0$ always.** From (b), L is positive semidefinite, so all $\lambda_k \geq 0$; and $L\mathbf{1} = 0$ shows 0 is attained with eigenvector $\mathbf{1}$. Thus $\lambda_1 = 0$ carries no information about the graph, which is why the second eigenvalue is used.

3 Theorem: $\lambda_2 > 0$ if and only if the graph is connected

Theorem 1. *The multiplicity of the eigenvalue 0 of L equals the number of connected components of the graph. In particular $\lambda_2 > 0 \iff$ the graph is connected.*

Proof. Suppose $L\mathbf{x} = 0$. Then $\mathbf{x}^\top L\mathbf{x} = 0$, and by (1) every edge has $x_i = x_j$. Following edges, $\mathbf{x}$ is constant on each connected component. Conversely, any vector constant on components is in the null space (each row of L only mixes a node with its neighbours, which share its value). So the null space is spanned by the indicator vectors of the components, and its dimension is the number of components c . Hence $\lambda_1 = \dots = \lambda_c = 0$ and $\lambda_{c+1} > 0$; with $c = 1$ this reads $\lambda_2 > 0$. $\square$

VERIFICATION 2 is a test of the implementation, not of the theorem: for every graph evaluated during the sweeps (87,000 evaluations; many are repeats, because the same placements are thresholded at different ranges) it computes both $[\lambda_2 > 10^{-9}]$ and a breadth-first search, and requires them to agree. Zero mismatches means the adjacency construction, the Laplacian, and the eigensolver are consistent with an independent algorithm. (The 10^{-9} tolerance absorbs floating-point round-off; λ_2 of a disconnected graph comes back as $\pm 10^{-15}$, not exactly 0.)

4 Two exact benchmarks

Proposition 1 (Complete graph). *For K_N (every pair linked), $\lambda_2 = N$.*

Proof. $A = \mathbf{1}\mathbf{1}^\top - I$ and $D = (N-1)I$, so $L = NI - \mathbf{1}\mathbf{1}^\top$. On the subspace $\mathbf{x} \perp \mathbf{1}$ we have $\mathbf{1}\mathbf{1}^\top \mathbf{x} = 0$, so $L\mathbf{x} = N\mathbf{x}$: the eigenvalue N has multiplicity $N-1$. Together with $\lambda_1 = 0$ this is the whole spectrum, and $\lambda_2 = N$. $\square$

VERIFICATION 1 checks this for $N = 10, 20, 30$. It catches a wrong sign convention for L or a wrong eigenvalue index.

Proposition 2 (Fiedler’s degree bound [1]). *Let $\delta_{\min} = \min_i d_i$. Then $\lambda_2 \leq \frac{N}{N-1} \delta_{\min}$.*

Proof. By the Courant–Fischer characterisation, $\lambda_2 = \min_{\mathbf{x} \perp \mathbf{1}, \mathbf{x} \neq 0} \mathbf{x}^\top L\mathbf{x} / \mathbf{x}^\top \mathbf{x}$, so any particular $\mathbf{x} \perp \mathbf{1}$ gives an upper bound. Let v be a node of minimum degree and take $\mathbf{x} = e_v - \frac{1}{N}\mathbf{1}$ (orthogonal to $\mathbf{1}$ since its entries sum to $1 - 1 = 0$). Every edge at v contributes $(x_v - x_j)^2 = 1$ and every other edge contributes 0, so $\mathbf{x}^\top L\mathbf{x} = d_v = \delta_{\min}$. Also $\mathbf{x}^\top \mathbf{x} = (1 - \frac{1}{N})^2 + (N-1)\frac{1}{N^2} = \frac{N-1}{N}$. Hence $\lambda_2 \leq \delta_{\min} \cdot \frac{N}{N-1}$. $\square$

Correction. The first draft of the script tested the bound $\lambda_2 \leq \delta_{\min}$ and failed on 16,195 graphs. The test was wrong, not the code: for K_N , $\lambda_2 = N$ exceeds $\delta_{\min} = N-1$. The bound above is the correct one and holds on all graphs (VERIFICATION 3).

5 Why λ_2 and not “number of live links”

Two graphs with the same number of links can differ completely: 20 links scattered evenly leave a network in one piece; 20 links concentrated on the only path between two clusters, if removed, split it. λ_2 reflects that structure. It also has a dynamical meaning: for the consensus dynamics $\dot{\mathbf{x}} = -L\mathbf{x}$ (each node moves toward the average of its neighbours), the disagreement decays as $e^{-\lambda_2 t}$ [2]. A falling λ_2 therefore means *slower* coordination before it means *no* coordination, which a binary connected/disconnected flag cannot show. For that reason the script records, besides the collapse threshold, the J/S at which the mean λ_2 has fallen to half its unjammed value (the “knee”).

6 Links as a distance threshold

The link range. In this model the jammer is far from all nodes, so every receiver sees the same J , while the received signal falls with link length d . Both $E_b/N_0(d)$ and $E_b/N_J(d)$ decrease with d , so the link BER increases with d , and

$$P_b(d_{ij}) \leq \text{BER}_{\text{out}} \iff d_{ij} \leq r(J/S, \rho),$$

where the *link range* r solves $P_b(r) = \text{BER}_{\text{out}}$. The link graph is therefore a random geometric graph [3]: N uniform points, joined when closer than r . VERIFICATION 5 confirms that the BER rule and the distance rule give identical adjacency matrices for every graph of the J/S sweep.

The jammer's choice of ρ . Removing edges cannot increase λ_2 : by (1), fewer edges make $\mathbf{x}^\top L \mathbf{x}$ smaller for every $\mathbf{x}$, hence also its minimum over $\mathbf{x} \perp \mathbf{1}$. A smaller range removes edges, so the ρ that minimises λ_2 is the ρ that minimises r . It is the same for every placement and every N , and the script computes it once per J/S value (`worst_rho()`). VERIFICATION 4 checks the monotonicity placement by placement; VERIFICATION 6 checks that the ρ with the lowest mean λ_2 on the grid of the ρ -sweep is the grid neighbour of the range-minimising ρ .

Which ρ that is (figure `s2_lambda2_vs_rho`). Minimising the range means bringing down the strongest possible link, which is the question answered in Section 4 of the S1 note:

$$\rho_{\text{out}} \approx \frac{\text{BER}_{\text{out}}}{Q(\sqrt{x^*})} = 8.55 \text{BER}_{\text{out}} = 0.0086.$$

The exact search gives 0.0091 at $J/S_{\text{ref}} = -14$ dB (thermal noise moves it slightly) and λ_2 is lowest at the nearest grid point, 0.0089. This is an order of magnitude below the 0.135 of the S1 figure, but for a simple reason: both are $\rho^* = 0.709/(E_b/N_J)$, evaluated at $E_b/N_J = 8$ dB in the S1 figure and at the outage point 19.2 dB here. It is not a network effect, and it scales with the assumed criterion ($\text{BER}_{\text{out}} = 10^{-2}$ would give $\rho_{\text{out}} = 0.086$). The rise of λ_2 on the left of the figure has a separate cause: a jammed hop has a BER of at most $\frac{1}{2}$, so $P_b \leq \rho/2$ plus the thermal term, and no jammer power can bring a link down once $\rho < 2 \text{BER}_{\text{out}}$.

One-channel limit. $\rho_{\text{out}} = 0.0086$ is just below $1/N_{\text{ch}} = 0.01$, the smallest fraction a jammer can occupy with 100 channels; the figure shades that region. Enforcing $\rho \geq 0.01$ changes the range at -14 dB from 538.8 m to 539.7 m and the thresholds below by 0.03 dB.

7 The range–power relation and the collapse threshold

Closed form. When jamming dominates thermal noise, the longest surviving link has $E_b/N_J = 0.0829/\text{BER}_{\text{out}}$ (S1 note, Section 4). With $E_b/N_J(d) = G_p - J/S_{\text{ref}} - 10n \log_{10}(d/d_{\text{ref}})$ in dB,

$$J/S_{\text{ref}}(r) [\text{dB}] \approx G_p - 10 \log_{10} \frac{0.0829}{\text{BER}_{\text{out}}} - 10n \log_{10} \frac{r}{d_{\text{ref}}}. \quad (2)$$

Three consequences:

- every threshold moves by +10 dB per tenfold increase of N_{ch} (as long as $\rho_{\text{out}} \geq 1/N_{\text{ch}}$), and by -10 dB per tenfold *decrease* of BER_{out} ;
- the *spacing* between the curves for different N is $10n \log_{10}(r_{N_1}/r_{N_2})$: independent of BER_{out} , G_p and d_{ref} , but proportional to the path-loss exponent n ;
- the absolute values of J/S_{ref} carry no meaning beyond the chosen d_{ref} ; the geometry enters only through r/L .

Collapse threshold (figure s2_lambda2_vs_js). J/S^* is defined as the J/S_{ref} at which half of the random placements are still connected. The script sweeps J/S_{ref} in 1 dB steps over 500 placements per N , interpolates the 0.5 crossing linearly, and obtains a 95 % interval by resampling the placements (bootstrap, 1000 resamples).

N	J/S^* [dB]	95 % interval	range at J/S^*	Eq. (2)	50 % knee
10	−11.79	−11.99 .. − 11.61	$0.421 L$	−11.68	−17.35
20	−9.42	−9.55 .. − 9.29	$0.323 L$	−9.36	−17.22
30	−7.92	−8.08 .. − 7.80	$0.272 L$	−7.88	−17.18

Going from 10 to 20 nodes buys 2.4 dB of jamming tolerance and from 20 to 30 another 1.5 dB: the gain per added node shrinks, because the range needed to keep N random points connected shrinks more slowly as N grows. Eq. (2) reproduces the simulated thresholds within 0.11 dB (VERIFICATION 7); the remainder is thermal noise. The knee, in contrast, is almost the same for all three sizes (−17.2 to −17.4 dB).

Limits of the model. Both results rest on the hard outage threshold without coding. With a coded packet-error model ρ_{out} and the threshold values would change; with a jammer at a finite distance the received J would differ between nodes and the graph would no longer be a plain random geometric graph. Both are listed as extensions in the README.

8 Correspondence with the code

Mathematics	Code
$A_{ij} = [P_b \leq \text{BER}_{\text{out}}]$	<code>adjacency()</code>
$L = D - A$	<code>lap = np.diag(a.sum(axis=1)) - a</code>
λ_2 (2nd smallest, symmetric)	<code>np.linalg.eigvalsh(lap)[1]</code>
connected? (independent)	<code>is_connected_bfs(a)</code>
link range $r(J/S, \rho)$	<code>link_range()</code>
range-minimising ρ	<code>worst_rho()</code>
Eq. (2)	<code>js_ref_closed_form()</code>
0.5 crossing, knee	<code>crossing()</code>
$K_N: \lambda_2 = N$	VERIFICATION 1
$\lambda_2 > 0 \iff$ connected	VERIFICATION 2
$\lambda_2 \leq \frac{N}{N-1} \delta_{\text{min}}$	VERIFICATION 3
λ_2 non-increasing in J/S	VERIFICATION 4
BER rule = distance rule	VERIFICATION 5
λ_2 -minimising ρ = range-minimising ρ	VERIFICATION 6
Eq. (2) vs simulation	VERIFICATION 7

References

- [1] M. Fiedler, “Algebraic connectivity of graphs,” *Czechoslovak Mathematical Journal*, vol. 23, no. 2, pp. 298–305, 1973.
- [2] R. Olfati-Saber, J. A. Fax, and R. M. Murray, “Consensus and cooperation in networked multi-agent systems,” *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007.
- [3] M. Penrose, *Random Geometric Graphs*. Oxford University Press, 2003.